CLASS 03 / SEP 8 / U02

CSG CONTINUED
+ 3D PRINTING

GEOMETRY THAT SURVIVES CONTACT WITH A MACHINE

NOTHING DUE

P2b ISSUED TODAY

CSG, CONTINUED

Finish the graph before it goes anywhere near a printer.

EXPOSE THE INPUTS

Pick the two or three numbers that change the design.

Expose them. Name them.

Change one, predict the result, then look.

CHECK IT IS CLOSED

Every difference and intersection must leave a closed solid.

Look for open edges, stray faces, parts that only touch at an edge.

Fix it in the graph, not on the mesh.

FINISH THE P2a DIAGRAM

One frame per operation, same view, one verb each.

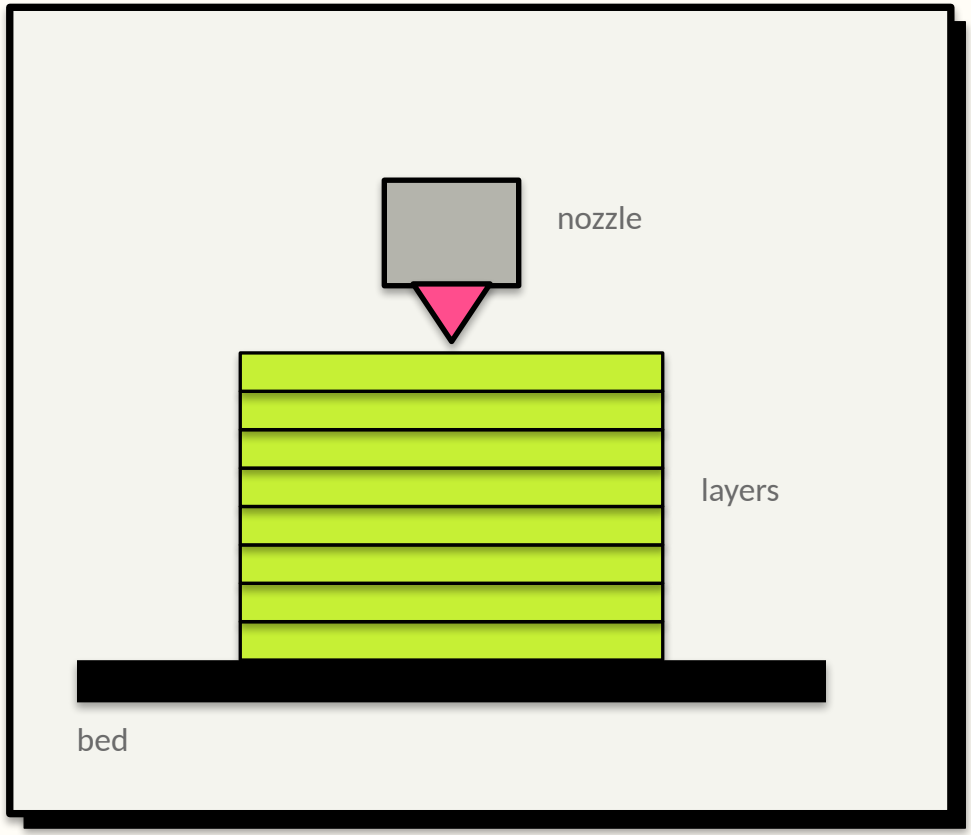
Mark the kept and the removed solid.

Draw the last frame at building scale.

ONE GRAPH, TWO ASSIGNMENTS: P2a DRAWS IT, P2b PRINTS IT. KEEP THE HISTORY INTACT.

WHAT THE MACHINE DOES

One thing, repeatedly. Everything else follows from this.



01

IT BUILDS UPWARD, NEVER SIDEWAYS

Material can only go on material that is already there.

02

LAYERS ARE THE WEAK DIRECTION

A print snaps between layers long before it snaps across one.

03

EVERY DECISION IS A TIME COST

Finer layers, more material, more supports: all of it is hours on a shared machine.

A NOZZLE DRAWS ONE FLAT LAYER. THE BED DROPS. IT DRAWS THE NEXT.

MANIFOLD OR IT DOES NOT PRINT

A model can look perfect on screen and mean nothing to a slicer.

WATERTIGHT

No gaps. Every edge is shared by exactly two faces. If water could leak out, it is not a solid.

NO SELF-INTERSECTION

A shape that passes through itself has no clear inside. The slicer cannot decide what to fill.

NORMALS OUTWARD

Every face has a front and a back. A flipped face tells the slicer that outside is inside.

CHECK BEFORE YOU SLICE. A BOOLEAN OPERATION THAT LEFT AN OPEN EDGE IS THE MOST COMMON CAUSE OF A FAILED FILE.

This is the closure check from the CSG slide, seen from the machine's side.

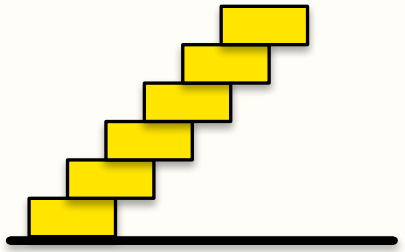
OVERHANGS AND THE LETTER TEST

Each layer needs something under it. How much is the whole question.



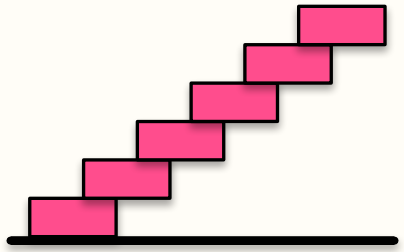
VERTICAL

Every layer sits fully on the one below.



45°

Each layer steps out as far as it is tall. About the limit.



STEEP

Most of the layer hangs in air. It droops or fails.

H — BRIDGE

PRINTS FINE

The crossbar spans two legs that already exist. Anchored at both ends.

V — SLOPE

FINE TO 45°

Each layer steps out a little. Past about 45° it starts to droop.

T — CANTILEVER

DROOPS

The arms start in mid-air, anchored at one end only.

Y — FORK

CHECK THE ANGLE

Two slopes grow from one stem. Steep arms print; flat arms droop.

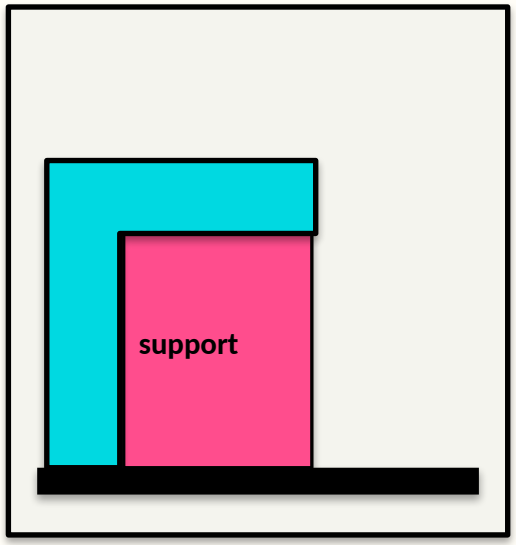
THE 45° RULE: PAST ABOUT 45° FROM VERTICAL, A SURFACE NEEDS SUPPORT — OR A DIFFERENT ORIENTATION.

An island, a part that starts with nothing under it, cannot print without support. Find all four conditions in your own model.

ORIENTATION IS YOUR ONLY LEVER

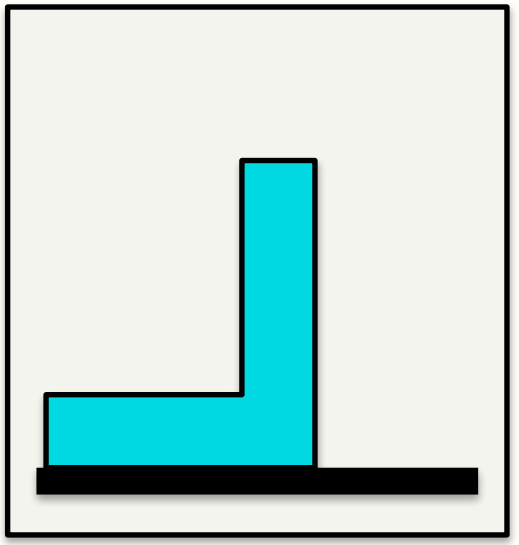
Same geometry. Two ways to place it. Very different prints.

AS MODELLED



The arm overhangs. It needs support, which costs material and time, and scars the surface where it is removed.

ROTATED



The arm now lies on the bed. No support, faster, a cleaner face, and stronger: the load no longer runs across a layer line.

NO SUPPORTS ARE ALLOWED ON P2b. IF NO ORIENTATION WORKS, CHANGE THE GEOMETRY AND EXPLAIN THE CHANGE ON THE PROCESS SHEET.

WHAT THE SLICER DOES

It turns a solid into a list of moves. You will do the same by hand in Week 12, Discretizing Geometry.

01 CUT INTO LAYERS

A stack of horizontal planes cuts the solid.

02 OUTLINE EACH LAYER

Each cut becomes a closed perimeter.

03 FILL THE INSIDE

An infill pattern, usually sparse.

04 ORDER THE MOVES

Perimeters, infill, travel, written as G-code.

FOUR SETTINGS — LEAVE EVERYTHING ELSE AT DEFAULT

LAYER HEIGHT

0.2–0.3 mm

Finer looks better and takes longer. Draft is fine here.

INFILL

10–40%

A lattice inside a shell. Solid is almost never needed.

WALLS

2–3

Perimeters, not infill, make a print feel solid.

SUPPORTS

OFF

Not allowed on P2b. Solve it with orientation.

LIVE NOW — WE SLICE ONE OBJECT TOGETHER AND READ THE PREVIEW: LAYER COUNT, PRINT TIME, MATERIAL.

PRINT IT OR CUT IT

Printing is not the default. It is the expensive option.

CUT IT

- Prismatic: one profile, extruded
- Flat, or made of flat pieces
- Repeated many times
- Larger than your hand

Minutes per sheet. Many parts at once.

PRINT IT

- Doubly curved: a vault, a shell
- Undercuts a blade cannot reach
- Detail in all three axes at once
- Small and intricate

Hours per object. One at a time.

THE TEST: CAN IT BE DESCRIBED AS A STACK OF FLAT PROFILES, OR ONE PROFILE EXTRUDED? THEN CUT IT. A SIMPLE BOX, PRINTED, IS A BOX YOU WAITED HOURS FOR.

P2b is printed on purpose. The laser cutter comes back in Project 4.

SIX WAYS IT GOES WRONG

Learn to name the failure. Naming it is most of the diagnosis.

WARPING

Corners lift; the base curves.

FIX Clean bed, a brim, no drafts.

ELEPHANT'S FOOT

The bottom layers bulge out.

FIX Slicer setting, or chamfer the base.

STRINGING

Fine hairs between parts.

FIX Retraction on. Fewer travel moves.

LAYER SEPARATION

It splits along a layer line.

FIX Reorient so the load runs along layers.

DROOP

An overhang sags into the air.

FIX Reorient. If that fails, change the geometry.

SPAGHETTI

A nest of filament in mid-air.

FIX Adhesion. Watch the first layer.

FABLAB RULES

Finish the FabLab safety orientation first. No orientation, no machine.

Never touch the hot end. The nozzle runs at about 200–250 °C and the bed is hot.

Never restart someone else's print. If a job looks wrong, tell staff.

The queue is first come. A print that starts at the deadline does not finish.

Booking, file naming, pick-up and material allocation: follow the procedure posted by FabLab staff.

ISSUED TODAY

P2b — CSG 3D PRINTED OBJECT

DUE WEEK 10 — Oct 27 / 7%

Print a variant of the CSG object you are diagramming for P2a. Small, unsupported and documented: the orientation, the slicer settings and a photograph go on one process sheet.

SIZE	SUPPORTS	SHEET
60 mm, 200 g Within 60 mm on each axis, inside the 200 g allocation.	**NONE** Orientation is the lever. Change geometry only if you must, and say why.	**ONE PAGE** Orientation and a photograph. Handed in with the print in Week 10.

REQUIREMENTS — P2b

Visual documentation 20% · computational understanding 20% · technical execution 15% · fabrication quality 30% · experimentation 5% · requirements and identity 10%.

01	Print a variant of your CSG object from P2a, within 60 mm on each axis and the 200 g allocation, without supports.
02	Hand in the print in Week 10 with one process sheet showing orientation and a photograph.
03	Submit the editable source, exported mesh and slicer project. Explain any geometry changes made for printability.

CHECKPOINT · WEEK 6 — Sep 29 — Check geometry, units, orientation and the sliced preview before queueing.

NOW — PREPARE YOUR PRINT

Rest of the class: take your CSG object from graph to sliced file, with help in the room.

✓	Solid checked — closed, no self-intersection, normals outward
✓	Scaled to fit within 60 mm on each axis, units confirmed in the slicer
✓	Oriented — no overhang past about 45°, supports off
✓	Sliced — layer count, print time and material written down
✓	Orientation, settings and any geometry change noted for the P2b sheet

LEAVE WITH A SLICED FILE AND ITS NUMBERS WRITTEN DOWN.

BEFORE NEXT CLASS

NEXT	WEEK 4 — Sep 15. Descriptive Geometry: point, line, edge, face, solid, Boolean operation. P3a is issued.
DUE	P2a, the CSG process sheet, is due next class, WEEK 4 — Sep 15.
PRINT	P2b print checkpoint WEEK 6 — Sep 29: Check geometry, units, orientation and the sliced preview before queueing. P2b is due at the mid-semester deadline, WEEK 10 — Oct 27.
FABLAB	Complete the FabLab safety orientation before the checkpoint. No orientation, no machine access.